Executive Summary

The following report outlines the purpose and the outcome of the project such as work completed to date, its key findings as well as discusses the prospects of the project. The purpose of the report is to serve as a reference guideline to designing and building a similar recirculating water pilot plant, including examples of instrumentation calibration.

In this project, a recirculating water pilot plant was built out of various piping materials and included several units and instruments. In order to build a plant, the project has gone through four phases; Phase 1: Planning, Phase 2: Design, Phase 3: Construction, and Phase 4: Calibration & Operation.

In phase 1: Planning, safety training was completed. The group was created, and the project roles were assigned.

In phase 2: Design, based on the pilot plant requirements and available pipe materials, fittings and instrumentation, the recirculating water pilot plant was designed and preliminary-assembled in lab. The design was outlined as a PFD and later on P&ID.

In Phase 3: Construction, the permanent assembly of the pilot plant was completed according to the drawn PFD and P&ID. The budget of the project was assessed via Bill of Lading and totalled 1279.15$. Multiple water leaks were detected during the first plant startup and eliminated by adding more Teflon tape around the threaded joint.

In Phase 4: Calibration & Operation, the assembled pilot plant began its operation. In this phase, startup, operation and shutdown manual were developed. As it was anticipated, due to the height of the plant, the pump had a high static head a required a lot of pump

work to bring the plant to its maximum flowrate. To aid the pressure loss within the system, the valve GL-101 was closed multiple times to build up the pressure around the pump. The maximum calibrated flowrate of this plant was only 18.3 LPM.

After the built pilot plant had successfully pumped the water in a recirculating loop, with no water leak, the calibration phase of the plant instrumentation began.

The rotameter and pressure gauges were calibrated. The pressure gauges were calibrated outside of the plant, using various pipe heights and water density at a measured temperature.

The rotameter calibration was calibrated by adjusting valve GL-101 until the water level aligned with the flowrate marker on rotameter FI-01. The nominal flowrate does not significantly differ from the calibrated flowrate.

In the final weeks of the project, the group will be conducting the studies on the transport phenomena of fluid mechanics, specifically pressure losses. It will be conducted through measuring pressure loss using the gauges on the plant and comparing to a value calculated using the Darcy-Weisbach Equation. Equivalent length of the plant will be calculated. The studies on heat transfer will be conducted on overall heat transfer coefficient. This will be done by calculating heating tape capacity (Q) by determining the change in water temperature in the system when the tape is on.

Finally, the transport phenomena of fluid mechanics and heat transfer studies results will be presented in the summary presentation.

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Design Principles

The purpose of the plant was to pump water in a recirculating loop from a holding tank, through piping of various materials, instruments, and control devices, and then return the water back to the holding tank via a flexible hose. The pilot plant serves as a demo for study of fluid mechanics and heat transfer. The plant accommodates removal of water through several parts on the loop, sample line and drain line. To accommodate that, the sample line and drain line were added into the pilot plant design. The drain line was chosen to be right next to the holding tank, to, if needed, drain the tank more efficiently. Refer to figure 1 for material placement on the plant (Refer to figure 1.1).

Pipping Material:

PVC 1 ¼'' schedule 80 was picked as the material and diameter of the fittings because the out of the tank line required connection to multiple valves such as the out of the tank valve (GV-101) and drain line (GV-102). Ensuring that the line fittings are the same diameter as the chosen valves, reduces the amount of fittings used therefore reducing the total cost and pressure loss. PVC material near the tank (T-101) and pump (P-101) makes it easier to install a union fitting, that would allow for a smoother construction process.

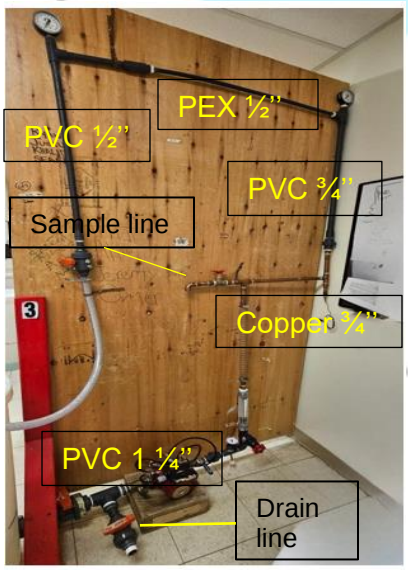
Copper ¾'' copper was used as a material in after the pump line because it was important not to lose pressure as the line pumps vertically due to the high static head of the pump. According to Engineering toolbox, Copper and the like metals have absolute roughness coefficient that ranges from 0.001-0.002 mm compared to PVC and other smooth plastic pipes absolute roughness coefficient of 0.0015-0.007 mm (2003). Therefore, copper induces less friction and therefore pressure/head loss.

PVC ¾ schedule 80 pipe was used on the after-pump line due to schedule 80 having a higher wall thickness ensuring that the pipe can withstand a pressure exerted as pump increases the pressure in the system. PVC ¾'' is easy to connect to copper ¾'' using minimum number of fittings, no reducers required as they are the same nominal diameter.

PEX ½'' SDR 9 allows to connect ½'' and ¾'' PVC pipes with minimal number of fittings. Crimpled joints ensures that the joints are leak proof.

PVC ½'' schedule 40 was used at the end of the loop, as the end of the loop has the lowest pressure within the system due to pressure loss, schedule 40 wall thickness stuffies on this line. The smaller diameter increases the friction coefficient and decreases the pressure. As the line is located at the end of the loop, before the tank and the line flow vertically down, alongside gravity force, the pressure loss does not matter significantly, as the flow is going down by gravity force regardless.

Figure 1.1. Pilot plant with labeled pipe material



Joints/Fittings:

Union installed after the tank before the valve (GV-101) allows for an easy tank (T-101) disconnection from the piping system as unit is heavy and if full of water, is impossible to move and avoid spillage (Refer to figure 1.2). The union installed near the pump allows to connect the pump (P-101) to the plant. This way if the pump needs replacement, it can be easily detached from the entire system and replaced without redoing any joints. To connect the pump to the system, the pump can stay on the floor and get attached to the already permanently joined piping system (Refer to figure 1.3).

Figure 1.2. Union at the tank



Figure 1.3. Union at the pump (P-101)



Instrumentation:

Refer to Figure 1.4. for valve placements.

Globe Valve

Rotameter PVC globe valve (GL-101) globe valve located before the rotameter used to adjusts the flow coming into rotameter. PVC was picked as a material to connect to PVC line with minimum fittings.

- Sample line brass globe valve (GL-102) open/close sample line adjusts the flow of the sample line according to the need. Located at the very end of sample line for an easy drain.

Gate Valve

- Out of tank gate valve (GV-101) open/close to immediately shut down or start up water flow out of the tank during startup/shutdown procedure of the plant. Does not require flow control.
- Drain line gate valve (GV-102) open/close drain line during startup to bleed off the system, located at the bottom next to the tank to drain the system during shutdown and easily drain the tank. Flow control is not necessary for drain valve; gate valve was picked to reduce the cost.
- Back to the tank gate valve (GV-103) used in system startup and shutdown to build up a pressure and stop the water coming into the tank in case of an emergency. No water flow control required.

Pump (P-101) was installed on the bottom due to its weight and at the beginning of recirculating plant, before the vertical lift to pump the water up (refer to figure 1.5).

HE-101 is located on the copper pipe due to metal heat conductivity. Its located farther away from the PVC pipe to avoid plastic pipe melting in case of overheating and farther away from thermocouple to ensure accurate flow temperature readings (Refer to figure 1.5).

Figure 1.4. Location of valves on the pilot plant

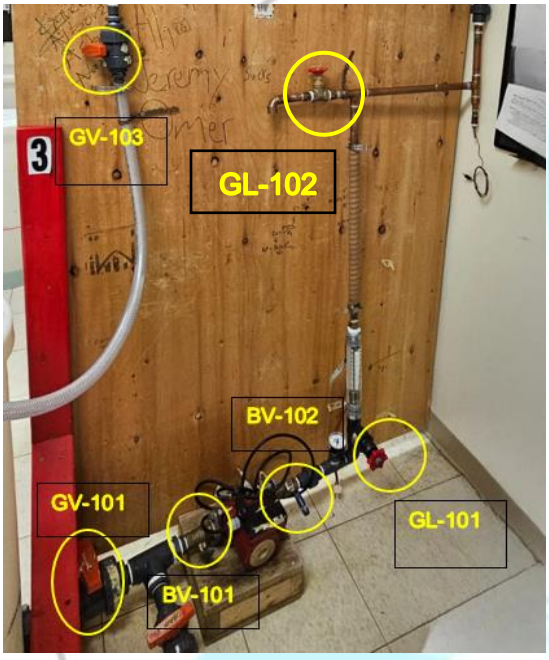
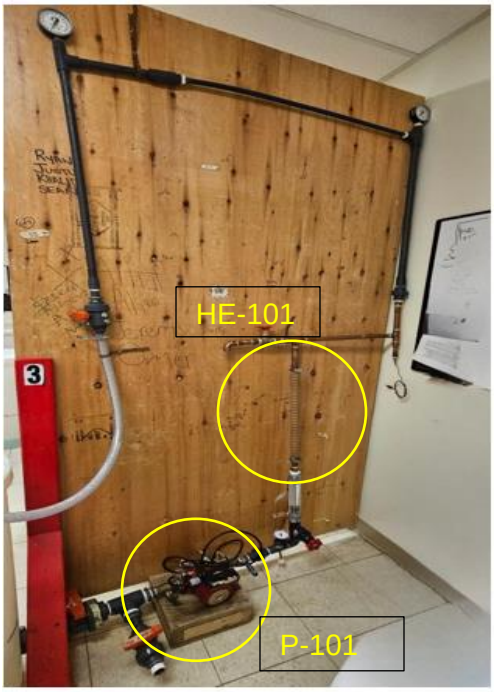


Figure 1.5. Pump and heating tape location on pilot plant



Sensors:

Refer to figure 1.6 for sensor placements on pilot plant.

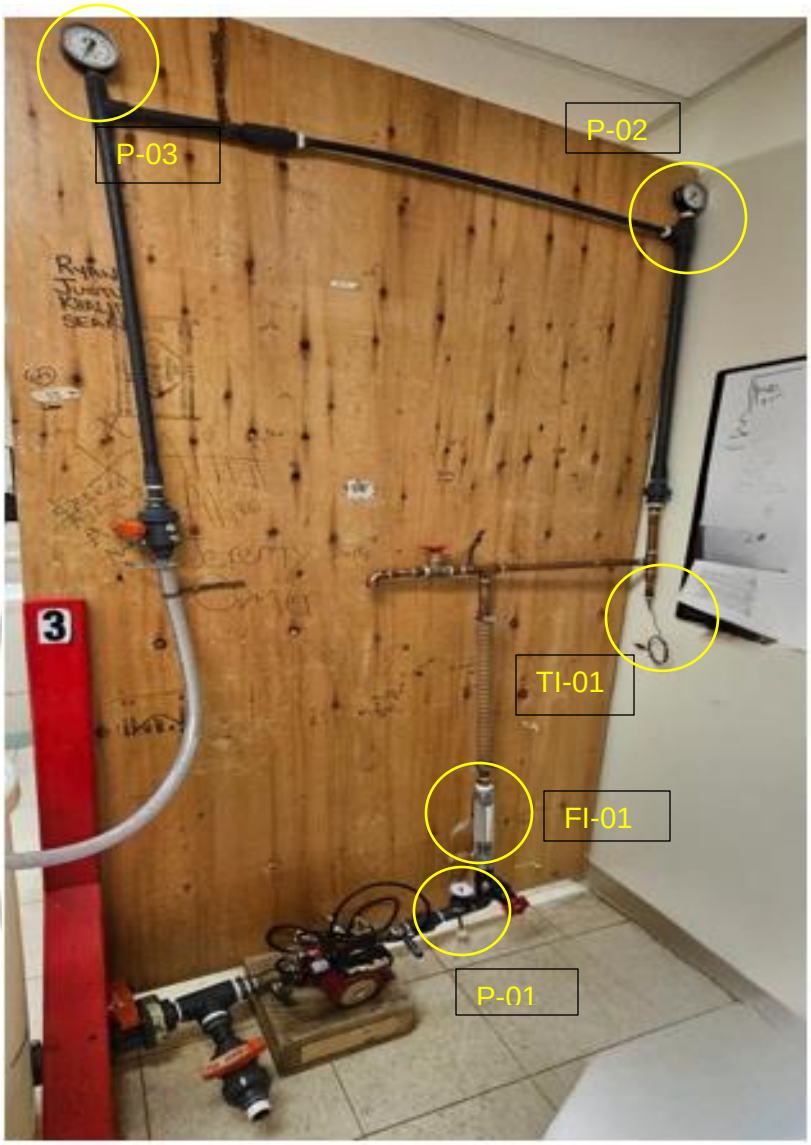
Rotameter (FI-01) located after the pump to minimize suction disturbances that could occur if placed before the pump.

Thermocouple TI-01 placed on copper pipping because it conducts the heat. Thermocouple is more accurate than a thermometer due to an extra significant digit in readings. It also is located farther away from heating tape to read flow temperature more accurately, rather than the temperature of the heated spot on copper as it is always going to be hotter.

Pressure Gauges were placed in three different parts of the plant to measure the pressure loss across the plant.

- Pressure gauge right after the pump (PI-01) it is known that the pump increases the pressure. Pressure gauge was placed right after the pump (P-101) to collect the reading of the initial, highest-pressure exerted by the pump.
- Pressure gauge on the top right (PI-02) was installed after the system travels upwards vertically to monitor pressure loss.
- Pressure gauge on top left (PI-03) monitors pressure of the system after it travels through a decreased diameter and close to completing the loop.

Figure 1.6. Sensors location on pilot plant



Start-up Procedure

The pilot plant that will be operated is ~80” tall and ~50” wide water heating system with a recirculating pump, storage tank, heating tape and Flow, Temperature and pressure indicating sensors.

Figure 2.1. The front and side view of pilot plant.



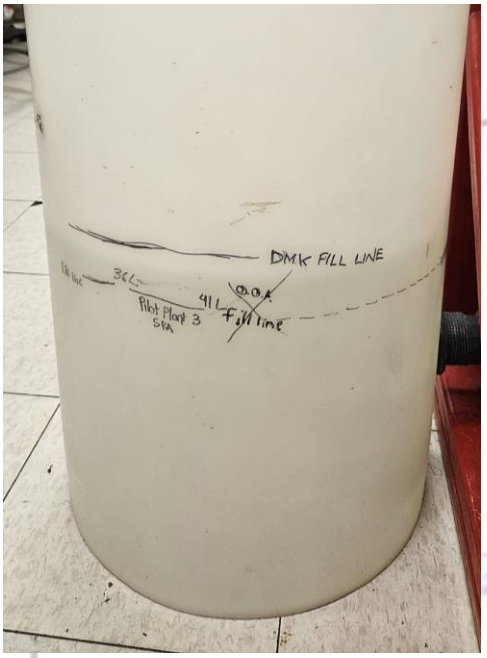
Table 2.1. Different parts and the names used in the startup and shutdown procedure.

Parts operated	Common name used	Name used in P&id
Valves	Tank Valve	GV-101
	Drain Valve	GV-102
	Pump Valve- Left	BV-101
	Pump Valve- Right	BV-102
	Globe Valve	GL-101
	Sample Line Valve	GL-102
	Back to Tank Valve	GV-103
Pressure Indicator	Pressure Gauge right after the pump	PI-01
	Pressure Gauge on top right	PI-02
	Pressure Gauge on top left	PI-03
Flow Indicator	Rotameter	FI-01
Temperature Indicator	Thermocouple	TI-01

The system can be started by following steps:

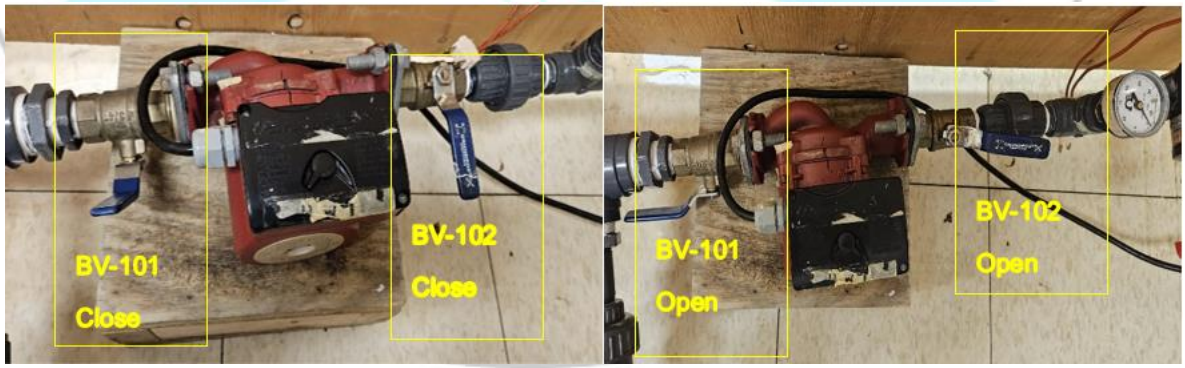
1. Fill up the tank till ‘DMK Fill Line’, by using the hose from the basin directly to the tank and ensure it is 40% full (40L water).

Figure 2.2. the tank foe pilot plant 3 and the DMK line represents the level of water to be filled up.



2. Open the pump valves, *BV-101* and *BV-102* 100%, by turning it to counterclockwise direction.

Figure 2.3. Direction of turning on the Pump valves for the startup procedure



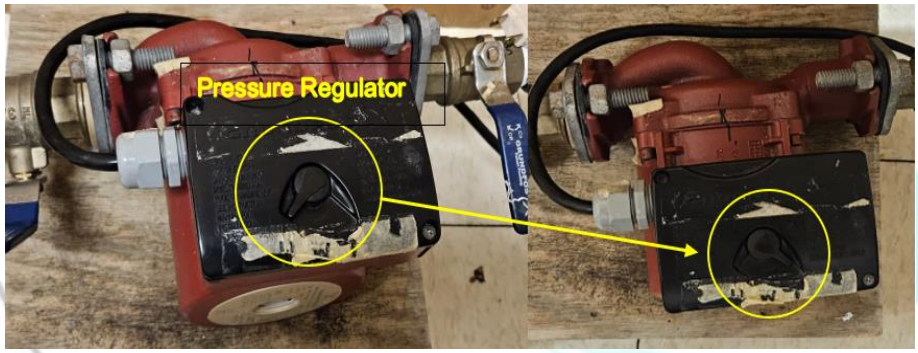
3. Open the tank valve, GV-101 by turning to counterclockwise, until it stops turning.

Figure 2.4. Tank valve to be opened.



4. Plug in the pump and run it at maximum pressure by adjusting the regulator slowly, from left to right.

Figure 2.5. Demonstration of regulating pump pressure.



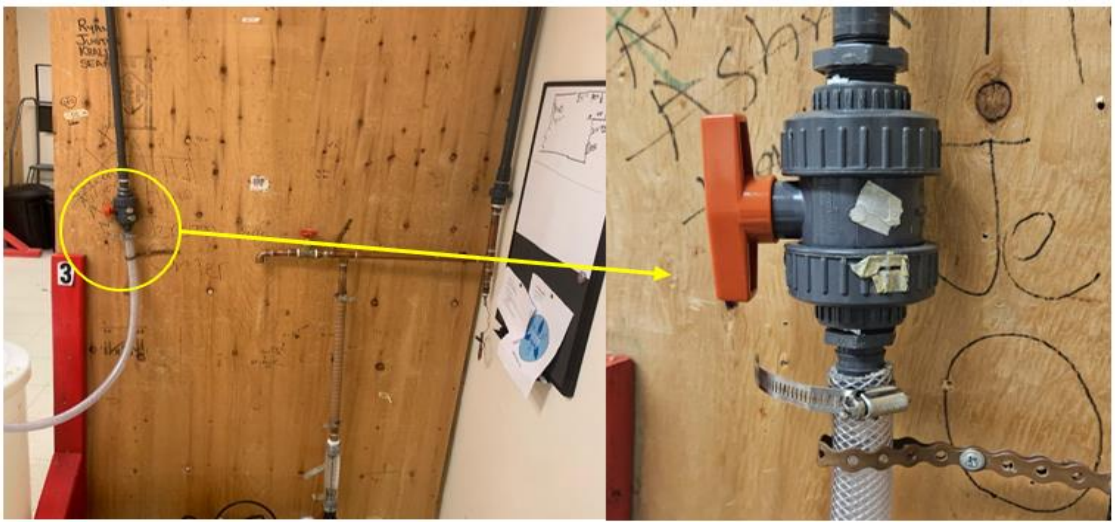
5. Open the Globe valve, *GL-101* by turning counterclockwise, until it stops turning.

Figure 2.6. The globe valve that is to be adjusted.



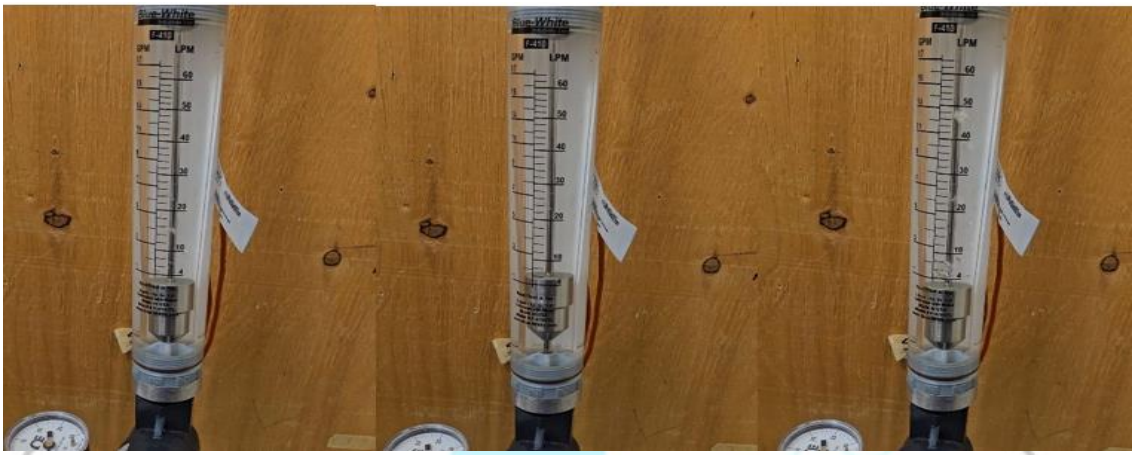
6. Build up the pressure by closing and opening the ‘back-to-tank’ valve, *GV-103* to release the air in the system. Turn clockwise and counterclockwise after every 2 seconds.

Figure 2.7. Back to tank valve which will be manipulated by opening and closing.



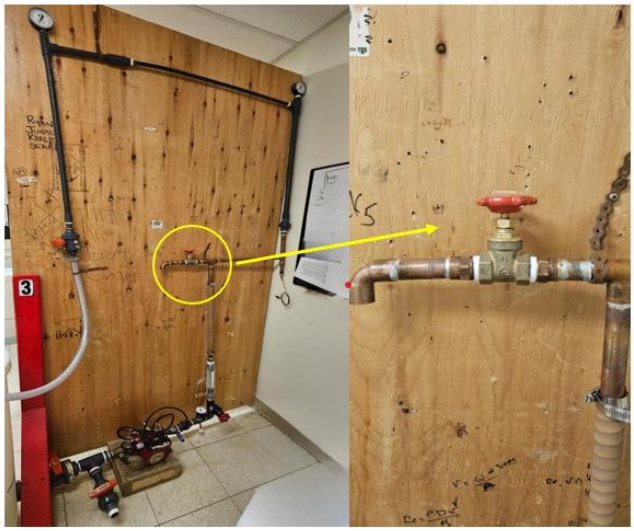
7. Take notice of the bubbles released in the rotameter, *FI-01* as well as the back-to-tank hose.

Figure 2.8. The float rises as the bubbles flow.



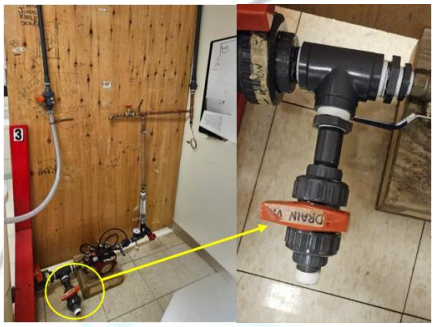
8. Continue adjusting the ‘back-to- tank’ valve *GV-103* by closing and opening continuously every 2 seconds for ~ 2 min.
9. After 2 minutes, open the ‘Sample Line’ valve, *GL-102*, by turning counterclockwise direction, for ~10 seconds to release the air stuck upstream and close it back, once done. Collect the water drained from the sample line into a bucket and dump it back to the tank.

Figure 2. 9. Sample Line valve that will be opened.



10. Open the ‘Drain’ valve, *GV-102* by turning counterclockwise for another 5-6 seconds to release the air stuck downstream, collect the drained water into a bucket and close it back once drained.

Figure 2.10. The drain valve of the plant.



11. Pour the drained water back into the tank to maintain the level of the tank, *T-101*.
12. Continue adjusting the ‘Back-to-Tank’ valve, *GV-103* as step 6.
13. Meantime with the help of another person, turn Glove Valve GL-101 clockwise until it stops turning to close the valve. Wait for 3-4 seconds to build up pressure and then turn it counterclockwise until it stops to completely open it. Repeat the adjustment of valve for around another 4 minutes.
14. After ~6 min, the float in the rotameter, *FI-01* starts to rise indicating the flow rate of the stream.

Figure 2.11. The float rises when the flow is stabilized.



15. Once the flow is stabilized, plug in the heating tape (the red wire) to heat up the water.

Figure 2.12. The heating tape of the pilot plant and the respective pug (red, which is plugged out in the picture) to turn it on.





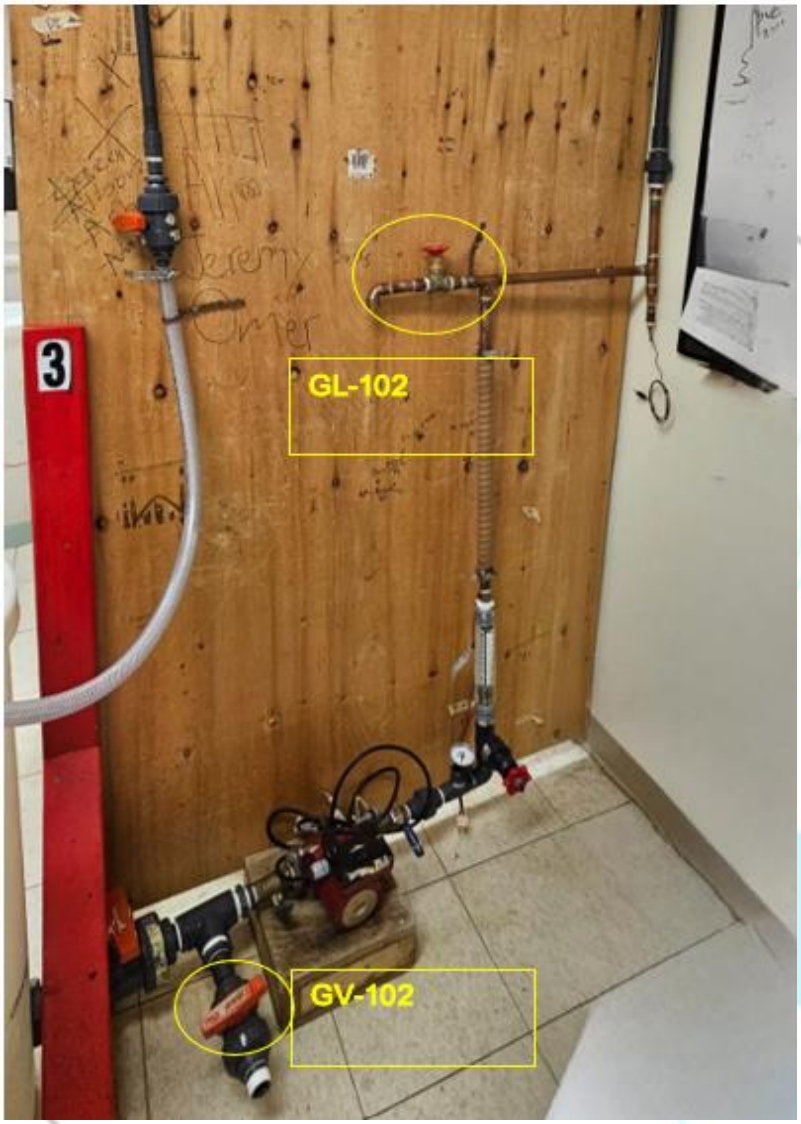
16. Wait for ~ 20 minutes to heat up the whole water in the tank.

Operation Guide

Safety guidelines to know before operating the system.

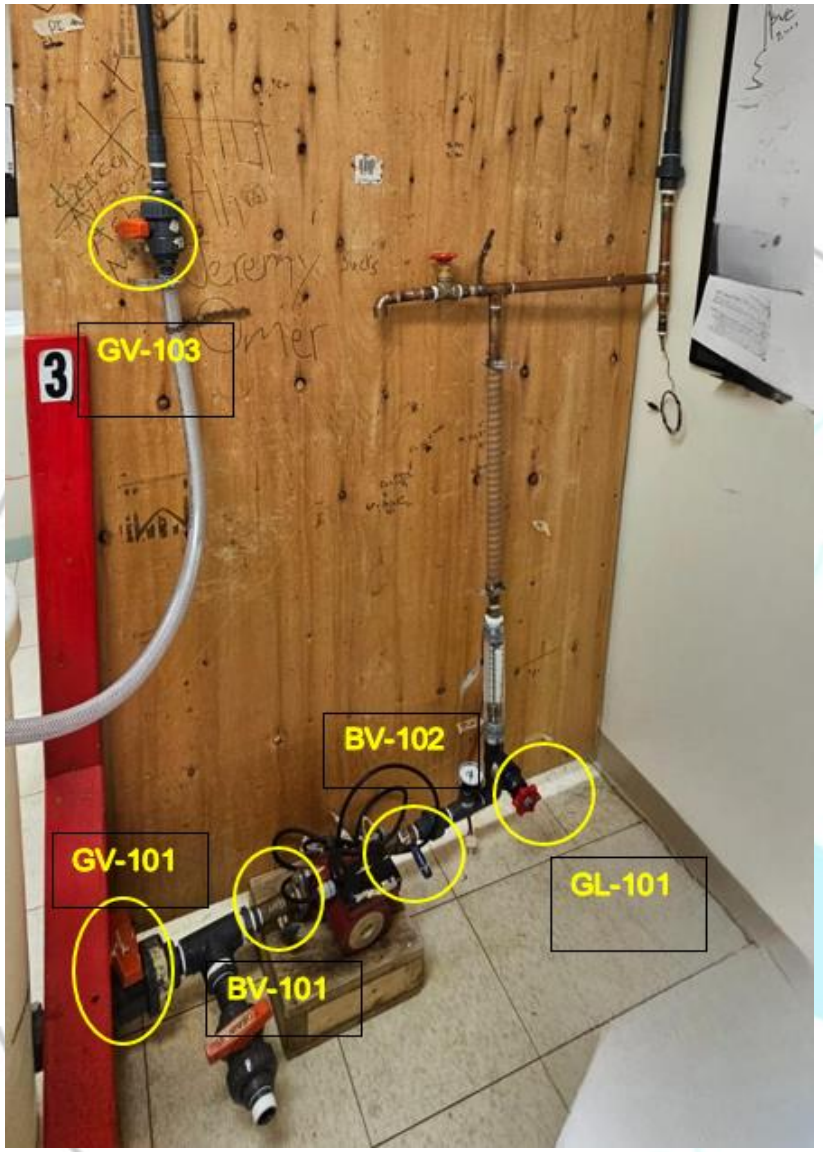
1. Make sure the pump is not running without any liquid flowing through.
2. Make sure the water is flowing continuously where the heating is done.
3. Adjust the flow rate of the whole system by adjusting the globe valve, *GL-101*.
4. Make sure not to touch heating tapes to avoid physical burns as well as any adjustments or air gaps between each loop of the tape.
5. Make sure the drainage and sample line valves (*GV-102 and GL-102*) are closed 100% always while operating unless it is necessary. Turn the valves clockwise until it stops, to close the valves.

Figure 3.1. The valves that are closed at the normal operating conditions.



6. Make sure the valves '*GV-101, BV-101, BV-102, GL-101 and GV-103*, are open 100% while operating unless it is necessary to close. Turn the valves counterclockwise to open fully.

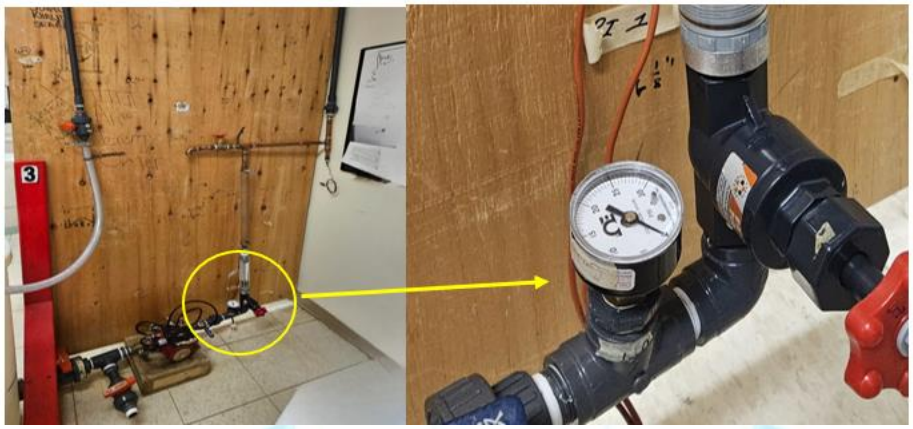
Figure 3.2. The valves that re supposed to be open all the time.



Operating the system

1. The system will take around 20 min to heat up the whole water present in the tank.
2. Note the pressure indicated by the pressure gauge, *PI-01* to know the pressure exerted by the pump, *P-101.*

Figure 3.3. The pressure exerted by the pump can be read on the pressure gauge.



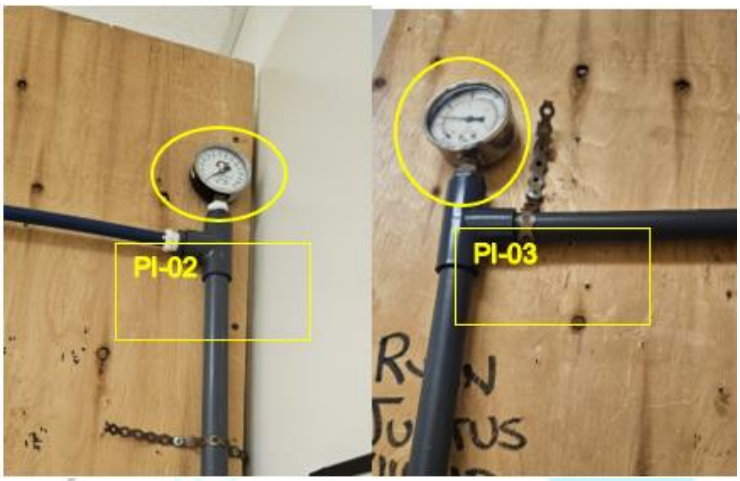
3. Note the flow rate indicated by the rotameter, F-01
4. Connect the thermocouple to a thermometer to take the temperature readings.

Figure 3.4. The temperature of the water can be measured by connecting thermometer to the thermocouple as given.



5. Wait until the water is heated to around 30^0C.
6. Take readings of the temperature to measure the rate of heating.
7. Note down the pressure readings from the Pressure Gauge, PI-02 and PI-03 respectively to measure the pressure drop in the system.

Figure 3.5. The pressure difference at 2 different points with 2 different pipe sizes, opposite flow direction with respect to gravity and at the maximum height of the plant.



Normal Operating Conditions:

After 17 minutes of startup, the following are the readings:

Table 3.1. The readings of the sensors after 15 min (including time for startup).

Sensor	Range	Reading
PI-01	0-30 psi	7.5 psi
PI-02	0-15 psi	0 psi
PI-03	0-15 psi	1.5 psi
FI-01	4-64 LPM	18 LPM
TI-01	-50°C - +1370°C **	22.9 ^{0}C

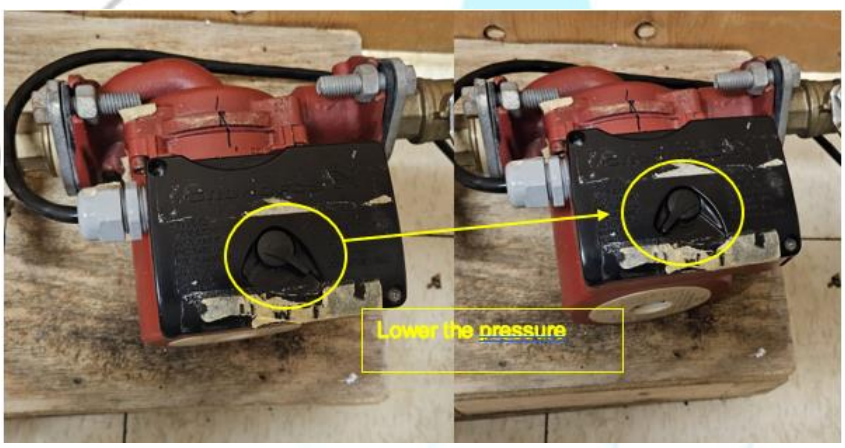
** Data deprived from Newark (n.d.). https://www.newark.com/omega/hh501ajk/temp-thermometer-50-to-1370deg/dp/68AK1225

Normal Shutdown Procedure

The Plant can be shut down through the following steps:

1. Plug out the heating tape.
2. Turn the pump back to minimum pressure, by adjusting the regulator slowly.

Figure 4.1. Demonstration of adjusting pump for shutdown.



3. Plug out the pump plug.
4. Close the tank valve, *GV-101,* by turning clockwise.
5. Open the sample line valve, *GL-102*, by turning anticlockwise to drain all the water above the sample line into a bucket and dump the water.

Figure 4.2. The sampling line valve to be opened.



6. Once completely drained from the sample line, close the sample line valve, GL-102, by turning in a clockwise direction until it stops.
7. Open the drainage valve, *GV-102*, 50 %, by turning the valve to 45^0 anti-clockwise for to drain all the water (from ground to sample line (~25“high)).

Figure 4.3. The drain valve to be opened.



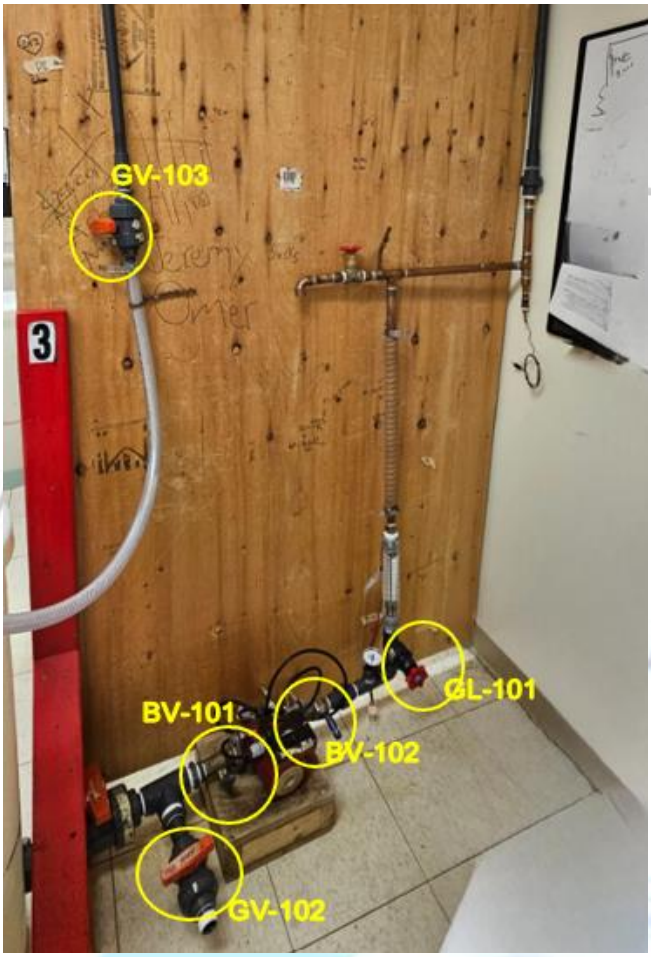
8. Adjust the drainage valve accordingly (closing and opening) so that when the bucket fills up, it can be emptied. Open the valve for 5 seconds, then close the valve by turning anticlockwise, until it is perpendicular to the pipe and then empty the bucket.
9. Continue draining as per steps 7 and 8 until it is completely drained, to avoid any leaking through the drain valve.
10. Check the float level in the rotameter and ensure it is dropped 100 % to the bottom of the instrument.

Figure 4.4. Rotameter after draining water from all the pipes.



11. Once the water is drained completely from the drain valve, close the drain valve, *GV-102*, Globe valve, *GL-101*, the pump valves, *BV-101 and BV-102* and the 'Back-to-Tank' valve, *GV-103* completely.

Figure 4.5. Valves to be closed for shutdown as per step 11.



12. Dump the bucket with drained water and empty it out.
13. Open the Tank valve, *GV-101*, and then drain valve, *GV-102*, slowly.

Figure 4.6. Valves to be opened to empty the Tank.



14. Empty the whole tank by slowly draining water into the bucket same as step 7 and 8. Make sure the drain valve is closed while dumping the water.
15. Once the tank, T-101, is completely drained, close tank valve*, GV-101* and the drain valve, *GV-102*
16. Shutdown completed.

Emergency Shutdown

The plant can be shut down as follows at times of emergency (or in cases of leaking):

1. Plug out the heating tape.
2. Turn the pump back to minimum pressure, by adjusting the regulator slowly.
3. Plug out the pump plug.
4. Close the tank valve, *GV-101*
5. Empty the pipelines by opening the sample line valve, *GL-102*, until the flow stops and close completely, once done.
6. Open the drain valve, *GV-102*, until the flow stops and close completely, once done.
7. Globe valve, *GL-101*, the pump valves, *BV-101 and BV-102* and the 'Back-to-Tank' valve, *GV-103* completely.
8. Dump the drained water back into the tank or washbasin accordingly.
9. Shutdown completed.

Instrument Calibration

Calibration is an important procedure of any project design that require the use of instrumentation. It is important to correct discrepancies between the indicated and measured measurements and improve the reliability of the data being recorded at any given reading.

Calibration of pressure gauges

The procedure that was used to calibrate pressure gauges PI-01, PI-02, and PI-03 involved measuring the pressure of tap water at different “water heights” within clear ½” PVC piping with a gauge attached elbow fittings to the farthest end. Hydrostatic pressure is induced within a water column and can be calculated as $P_h = \rho gh$, where ρ is the density of H_2O, g is the gravitational constant, and h is the water height within the tubing. For all three gauges, five independent measurements were taken to ensure accuracy, see **Appendix I**.

Figure 5.1. Location of PI-01, PI-02, and PI-03 on the pilot plant

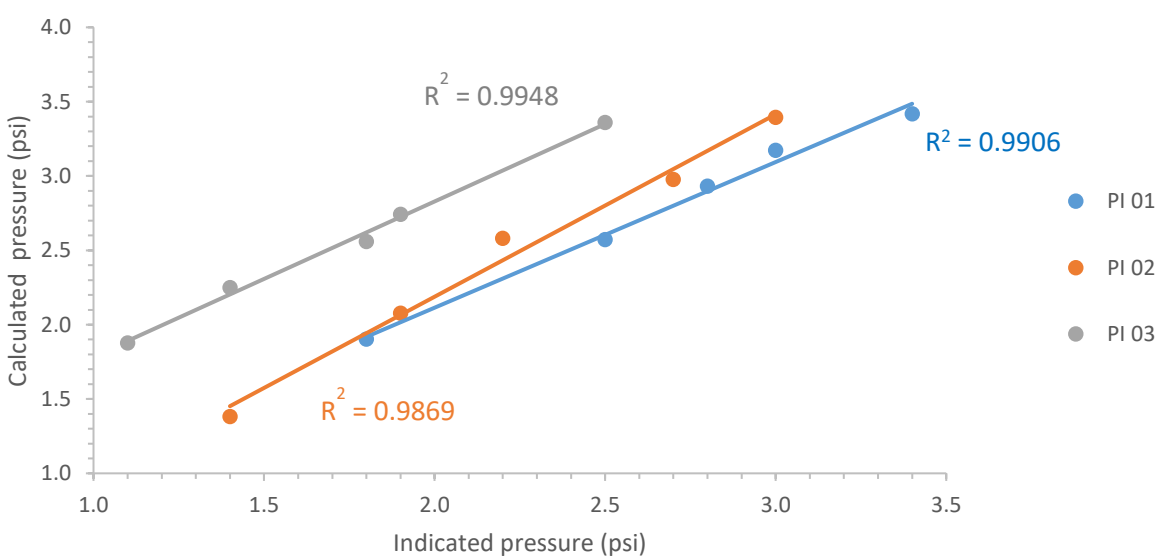


Offset between the indicated and measured pressure measurements may be as a result of but not limited to:

1. Mechanical Damage – physical damage to the gauge through shock or vibration can cause the misalignment of the internal components and cause permanent deformation influencing.

2. Temperature Changes – extreme temperatures can also contribute to the decrease in accuracy depending on the material of the internal components. Cheaper gauges tend to use cheaper materials that cannot withstand temperature variations, which can lead to the material expanding and contracting.
3. Corrosion – likewise, cheaper materials typically do not withstand prolong exposure to corrosive fluids (gases or liquids) and therefore will begin to degrade overtime.

Figure 5.2. Calculated P versus Indicated P (psi)



To calibrate the measurement taken on any of these instruments, use the following equations noting that variable ‘x’ represents the indicated pressure on the instrument and ‘y’ represents the calibrated measurement after calculation.

PI-01: y = 0.9799x + 0.1543

PI-02: y = 1.2269x – 0.2659

PI-03: y = 1.0419x + 0.7448

Calibration of Rotameter

The procedure that was used to calibrate the rotameter FI-01 involved measuring the time taken for the pilot plant to stabilize its flowrate, indicated by FI-01, and the volume of water collected within that time, measured using graduated cylinders, collected from the flexible reinforced hose connected to the tank. These measurements were done in triplicates for five different flowrates, see **Appendix II**. Flowrate can be described as the volume of fluid passing through a given cross-sectional area in a specified amount of time, $\dot{V}$ (volumetric flowrate) = V/t.

Figure 5.3. Location of FI-01 on the pilot plant

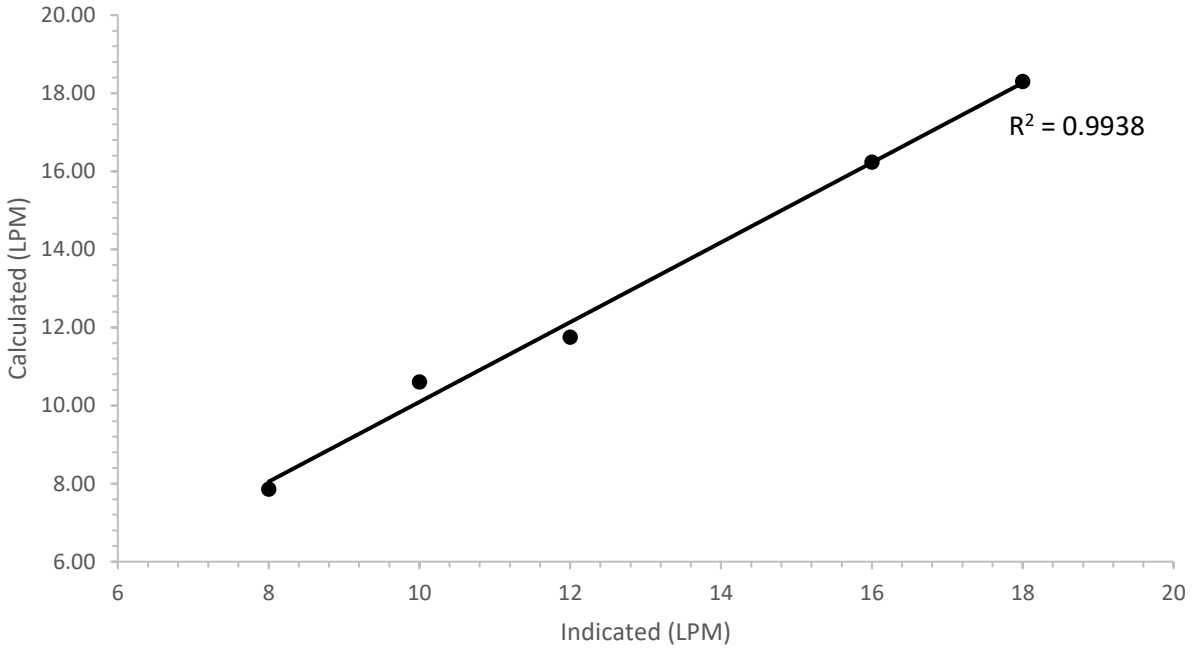


Offset between the indicated and measured flowrate measurements may be as a result of but not limited to:

1. Viscosity Changes – rotameters are usually calibrated for specific fluid viscosities and therefore if the fluid being measured deviates from the calibration viscosity, it can result in measurement inaccuracies.

2. Installation Errors – if there is a leak at the threaded joint of the rotameter because of improper alignment, this can cause flowrate deviation, or introduce tiny air pockets that can lead to the same error.
3. Instrument Age – likewise as instruments age, the internal components can degrade affecting the performance of the instrument. Components such as seals, O-rings etc. can wear out over time.

Figure 5.4. Calculated V̇ versus Indicated V̇ (LPM)



To calibrate the measurement taken on this instrument, use the following equations noting that variable ‘x’ represents the indicated pressure on the instrument and ‘y’ represents the calibrated measurement after calculation.

TI-01: y 1.0216x – 0.1245

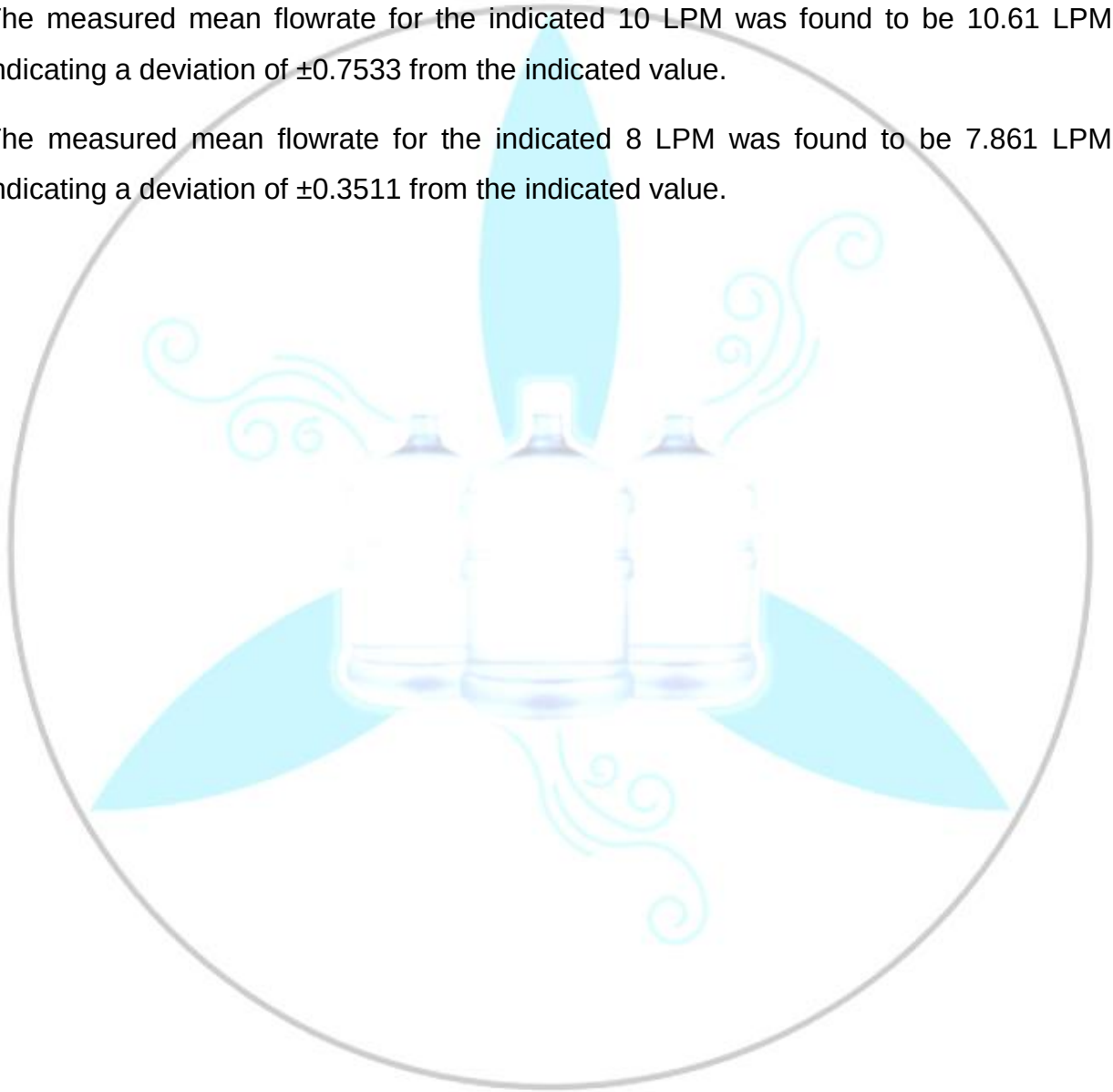
The measured mean flowrate for the indicated 18 LPM was found to be 18.30 LPM, indicating a deviation of ±0.2311 from the indicated value.

The measured mean flowrate for the indicated 16 LPM was found to be 16.24 LPM, indicating a deviation of ±0.6822 from the indicated value.

The measured mean flowrate for the indicated 12 LPM was found to be 11.75 LPM, indicating a deviation of ±0.3483 from the indicated value.

The measured mean flowrate for the indicated 10 LPM was found to be 10.61 LPM, indicating a deviation of ±0.7533 from the indicated value.

The measured mean flowrate for the indicated 8 LPM was found to be 7.861 LPM, indicating a deviation of ±0.3511 from the indicated value.



Appendix I

Calibration of pressure gauges

Table 4.1. Calibration of pressure gauge PI-01

Trial	TemperatureH_2O (°C)	Density H_2O (kg/m^3)	Pressure (psi)	H_2O height (m)	Pressure (psi)
1	27	996.53	1.8	1.34	1.9
2	26.4	996.69	2.5	1.82	2.6
3	26.3	996.71	2.8	2.07	2.9
4	26.2	996.74	3.0	2.24	3.2
5	25.9	996.82	3.4	2.41	3.4

Sample Calculation of Pressure:

$$P = ghp$$

$$P = 9.81\frac{m}{s^2} \times 1.34\, m \times 996.53\, \frac{kg}{m^3} = 1.9\, psi$$

Table 4.2. Calibration of pressure gauge PI-02

Trial	TemperatureH_2O (°C)	Density H_2O (kg/m^3)	Pressure (psi)	H_2O height (m)	Pressure (psi)
1	34	994.40	1.4	0.98	1.4
2	32.2	995.05	1.9	1.47	2.1
3	29.8	995.73	2.2	1.82	2.6
4	29.5	995.82	2.7	2.10	3.0
5	29.5	995.82	3.0	2.40	3.4

Table 4.2 Calibration of pressure gauge, PI-03

Trial	TemperatureH_2O (°C)	Density H_2O (kg/m^3)	Pressure (psi)	H_2O height (m)	Pressure (psi)
1	32.3	994.96	1.1	1.33	1.9
2	31.8	995.12	1.4	1.59	2.3
3	31.7	995.15	1.8	1.81	2.6
4	30.2	995.63	1.9	1.94	2.7
5	29.6	995.79	2.5	2.37	3.4

Appendix II

Calibration of rotameter

Table 5.1 Rotameter FI-01 calibration of nominal flowrate 18 LPM

Indicated (LPM)	Run	Time (s)	Volume (mL)	Calculated (LPM)	Mean (LPM)	Std Dev (±)
18	1	10.40	3210	18.52	18.30	0.2311
	2	10.30	3105	-		
	3	10.56	3225	18.32		
	4	10.20	3070	18.06		

Sample Calculation of Flowrate:

$$3210\ mL \times \frac{1\ L}{1000\ mL} \div (10.40\ s \times \frac{1\ min}{60\ s}) = 18.51\ LPM$$

Table 5.2. Rotameter FI-01 calibration of nominal flowrate 16 LPM

Indicated (LPM)	Run	Time (s)	Volume (mL)	Calculated (LPM)	Mean (LPM)	Std Dev (±)
16	1	10.65	2815	-	16.24	0.6822
	2	10.25	2881	16.86		
	3	10.15	2624	15.51		
	4	10.37	2824	16.34		

Table 5.3. Rotameter FI-01 calibration of nominal flowrate 12 LPM

Indicated (LPM)	Run	Time (s)	Volume (mL)	Calculated (LPM)	Mean (LPM)	Std Dev (±)
12	1	9.96	1905	11.47	11.75	0.3483
	2	9.94	2012	12.14		
	3	10.15	1970	11.65		

Data not shown under the **Calculated (LPM)** column indicated itself as an outlier within the measurements. The mean and standard deviation represent only the data that is shown.

Table 5.4. Rotameter FI-01 calibration of nominal flowrate 10 LPM

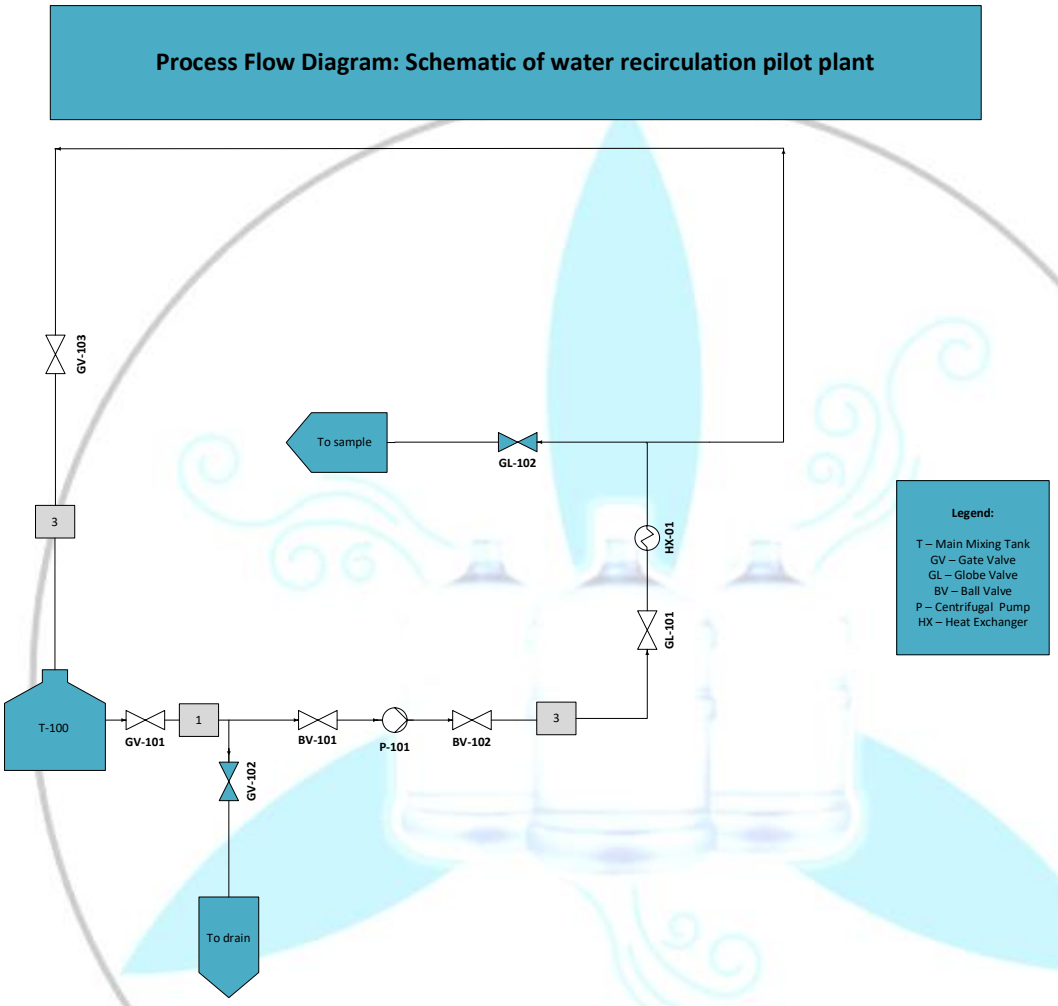
Indicated (LPM)	Run	Time (s)	Volume (mL)	Calculated (LPM)	Mean (LPM)	Std Dev (±)
10	1	10.15	1728	10.21	10.61	0.7533
	2	10.05	1697	10.13		
	3	10.30	1716	9.996		

Table 5.5. Rotameter FI-01 calibration of nominal flowrate 8 LPM

Indicated (LPM)	Run	Time (s)	Volume (mL)	Calculated (LPM)	Mean (LPM)	Std Dev (±)
8	1	10.40	1380	7.962	7.861	0.3511
	2	10.60	1446	8.185		
	3	10.20	1270	7.471		

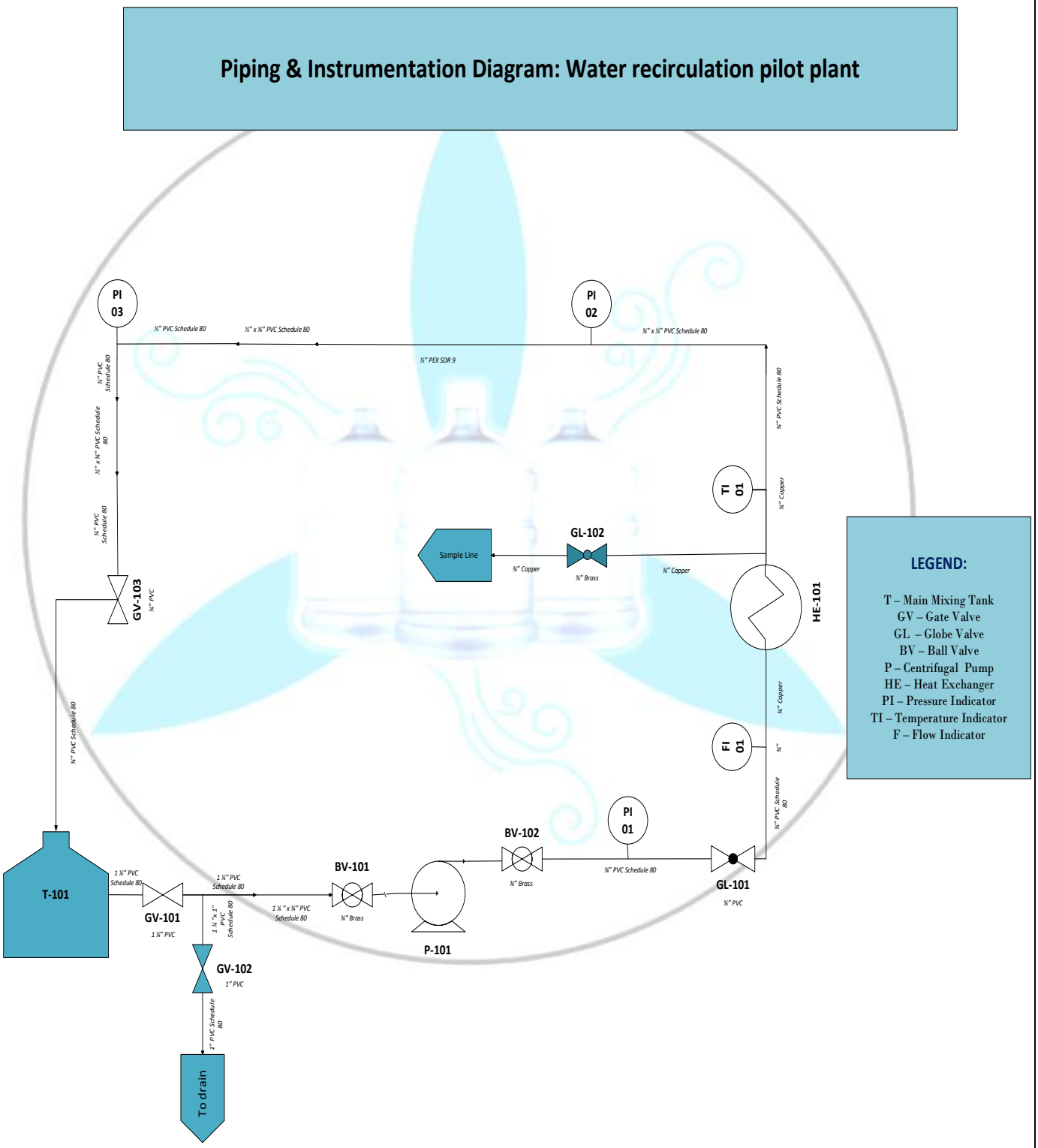
Appendix III

Figure 6.1. Process flow diagram



Appendix IV

Figure 7.1. Pipping and Instrumentation Diagram



Reference

The Engineering ToolBox (2003). *Roughness & Surface Coefficients* https://www.engineeringtoolbox.com/surface-roughness-ventilation-ducts-d_209.html

Newark (n.d.) *OMEGA HH501AJK Thermometer, -50°C to +1370°C, 192 mm, 91 mm, 53 mm, HH500 Series*. https://www.newark.com/omega/hh501ajk/temp-thermometer-50-to-1370deg/dp/68AK1225